

• It can carry large amounts of data.
Its applications include:

- Internet
- Cable TV
- Telephone
- Computer network
- Surgery and dentistry
- Lighting and decorations
- Mechanical inspection
- Military and space applications
- Automobile industry

Accurate testing of cables ensures that no good cables are rejected, or bad ones are accepted. Industries spend a lot of time and effort to develop better standards that ensure accurate testing. But these standards are generally written for manufacturers and not customers.

One of the measurements required to confirm the quality of installation is optical loss or insertion loss of each fibre in the cable. Loss of measurement is made on end-to-end basis in a permanently-installed cable plant. For this, a test source and optical power meter—sometimes called optical loss test set—and reference set cables are used. Sometimes, an optical time-domain reflectometer is also used.

The right tools for accurate optical-fibre cable testing

A lot of time and cost can be saved if the contractor and installers know the proper measurements that need to be done, how to make these measurements correctly, which tools to use and keep these tools in good condition—tools must be calibrated from time to time.

All errors that occur in an optical-fibre cable can be broadly classified into random errors and systematic errors.

Fibre attenuation measurement is essential. Attenuation is the loss of optical power as a result of absorption, scattering, bending and other loss mechanisms as light travels through fibre. Cut-back method is often used for measuring total attenuation of the optical-fibre cable. Modal bandwidth testing in a multi-mode optical-fibre cable is also essential.

Sometimes, chromatic dispersion occurs because index of refraction is a function of wavelength and different wavelengths of light travel through fibres at different speeds. Chromatic dispersion measurement can be made

INSTRUMENTS USED IN THE OPTICAL-FIBRE CABLE INDUSTRY

Test Parameter	Instruments used
Optical power (source output, receiver signal level)	Optical-fibre cable power meter
Attenuation loss of fibres, cables and connectors (insertion loss)	Optical power meter and source or optical loss test set
Source wavelength, spectrum width	Fibre-optical spectrum analyser
Backscatter for loss, length and fault location	Optical time-domain reflectometer
Fault location	Optical time-domain reflectometer, visual cable fault locator
Bandwidth dispersion	Dedicated bandwidth tester
Reflectance	Optical time-domain reflectometer, optical continuous-wave reflectometer
Geometry	Various mechanical and optical inspection tools

by wavelength selectable losers.

Optical-fibre cable geometry measurements are also important. For this, input end of the fibre is overfilled and mode-filtered. Output end of the fibre is viewed with a video camera. Image from the camera is sent to a computer, which analyses the same to identify the edges of core and cladding.

The following optical-fibre cable geometries are measured in the optical-fibre cable industry:

- Cladding diameter
- Core diameter
- Cladding non-circularity
- Core non-circularity
- Core-cladding non-circularity

Instruments used in the optical-fibre cable industry are summarised in the table above.

A submarine communication cable is a cable laid on the seabed between land-based stations to carry telecommunication signals across stretches of oceans and seas. Modern under-sea optical-fibre cables carry digital data, which include telephone, Internet and private data traffic.

AFL is a leading optical-fibre cable manufacturing company. Its cables were recently deployed through McMurdo ice shelf and into Ross Sea in Antarctica as a part of the study to measure the effects of warming of oceans on glacial ice. In such projects, frequent testing of optical-fibre cables is essential.

Majority of the Internet flows through under-sea cables and very little passes through satellite. India receives its data connections from three major landing ports, namely, Mumbai, Tamil Nadu and Kerala. Internet data reaches India through a network of under-sea

submarine optical-fibre cables owned by telecom giants such as Singtel, Etisalat, France Telecom and China Telecom. These cables are sometimes damaged by earthquakes, ship anchors and fishing trawls, and often need repair and maintenance. Repairing under-sea optical-fibre cables is like repairing punctured tyres.

Specifications of optical-fibre cables

Some factors that must be considered while selecting an optical-fibre cable are mechanical, environmental, dimensional characteristics with building codes and transmission equipment.

Normally, optical-fibre cables are classified into single-mode and multi-mode. However, these can also be categorised as outdoor, indoor and outdoor/indoor.

Optical-fibre cable specifications include attenuation and bandwidth, which are currently specified at two operating wavelengths for each fibre type. Performance is specified at 850nm, 1310nm and 1550nm for single-mode. Mechanical specifications include characteristics such as tensile strength, resistance to crush, impact and twist. Environmental considerations include temperature ranges for operation and storage. These also include sensitivity to moisture and sunlight, protection from lightning and rodent attack.

Commercially, two types of multi-mode optical-fibre cables are popular, each having a cladding diameter of 125 micron-metres but differing in core diameter. One has a core diameter of 50 micron-metres and the other of 62.5 micron-metres. These cables are called 50/125 and 62.5/125 micron-metre